Joint Spatial and Spectral Graph-Based Consistent Self-Representation for Unsupervised Hyperspectral Band Selection

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***Abstract*— Band selection (BS), which effectively reduces spectral dimensionality, stands out as a leading focus within hyperspectral image (HSI) analysis. Self-representation (SR) has surfaced as a favored technique in this domain due to its applicability to BS and unsupervised nature. However, the existing SR-based BS approaches only leverage either spatial or spectral relationships, with few integrating both while concentrating on the representation level rather than the selection level. In addition, employing all spatial pixels for spatial relationship utilization leads to considerable computational complexity. Therefore, this article proposes joint spatial and spectral graph-based consistent SR (JSSGCSR) to more effectively exploit spatial and spectral relationships for BS, which separately conducts SR to handle each view of spatial and spectral graphs to better consider two different structure characteristics, and ultimately integrates two SR results to achieve a unified and robust representative band set by imposing consistent sparsity pattern on their joint representation coefficients. In addition, the spatial and spectral relationships are integrated into different data spaces, that is, spectral graph SR and spatial graph SR are, respectively, conducted in the original HSI and the segmented and pooled HSI, which not only reduces the influence of superpixel segmentation on spectral relationships, but also improves the efficiency of spatial relationship utilization. Experimental results on three benchmark datasets have demonstrated the effectiveness of the proposed JSSGCSR in HSI classification tasks.**

***Index Terms*— Band selection (BS), hyperspectral images (HSI), self-representation (SR), spatial–spectral.**

I. Introduction

HYPERSPECTRAL imaging is an advanced technique used in remote sensing and various scientific applications to capture and analyze information from a wide range of electromagnetic wavelengths [1]. Unlike traditional imaging systems, hyperspectral imaging captures data across hundreds or even thousands of narrow, contiguous spectral bands. Each pixel in a hyperspectral image (HSI) contains a complete spectrum of information, allowing for a detailed analysis of the material composition of the imaged scene. Therefore, HSIs find widespread applications in disaster warning, environmental monitoring, precision agriculture, and resource exploration [2].

While hyperspectral imaging offers rich information, it also presents challenges in terms of data storage, processing, and analysis due to the large volume of spectral bands. First, its high-dimensional nature significantly reduces the efficiency of data transmission and increases computational workload. Second, the noise bands and interference bands generated during the imaging process diminish the accuracy of method analysis. Third, the high-dimensional feature classification is prone to the Hughes phenomenon [3], where the classification accuracy exhibits an initial increase followed by a decrease with the increase in feature dimensions. Therefore, to achieve rapid data transmission and effective processing, it is essential to perform dimensionality reduction on HSIs. As a crucial technique for HSI dimensionality reduction, band selection (BS) encompasses not only the essential data but also preserves the integrity of spectral features [4]. This is advantageous for enhancing the analysis efficiency of HSIs and has thus become a focal point in this research domain.

Based on whether labeling is required, BS can be broadly categorized into supervised and unsupervised methods [5]. Unsupervised methods, in comparison to supervised ones, have the advantage in that they do not require prelabeling and are not constrained by labeled data, making them more applicable to more types of datasets and tasks. Hence, this article explores the unsupervised category. Existing unsupervised BS methods mainly include the following categories. Ranking-based BS methods include information-based one which selects bands with higher information content [6], [7] and similarity-based one which adds bands sequentially that are least similar to the chosen set [8], [9]. Searching-based method formulates the BS problem as an optimization task with a specific objective function and utilizes exhaustive search, greedy search [10], [11], or heuristic search [12], [13], [14] to obtain the optimal band combination. Clustering-based BS methods group similar bands into clusters, and select bands corresponding to the center of each cluster as representative bands [15], [16], [17], [18]. Subspace partitioning-based BS methods first partition the HSI cube into multiple subcubes, then select one or more

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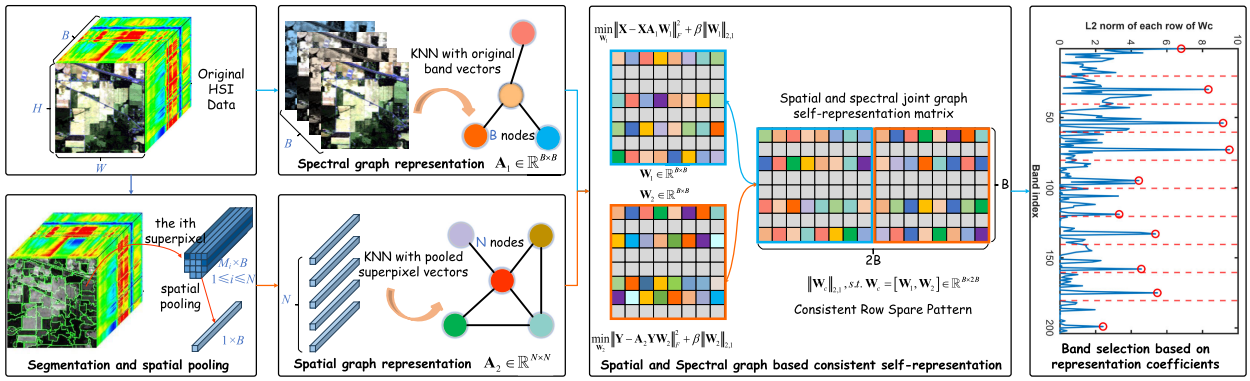


Fig. 1. Flowchart of the proposed JSSGCSR framework. First, spatial segmentation and pooling are conducted to obtain superpixels and its efficient representation, which aims to reduce the computational complexity of spatial relationship modeling. Second, the spectral and spatial graph is constructed to, respectively, capture the structure information at the level of spectral bands and spatial superpixels. Third, the spectral and spatial graphs are, respectively, integrated into an SR framework to explore and exploit the two kinds of characteristics, and a joint sparse regularization term is employed to maintain consistent sparsity and ensure consistent BS. Finally, representative bands can be selected based on the ℓ_2 norm of each row of the joint graph SR coefficient matrix.

bands from each subcube to represent the whole subcube [19], [20], [21], [22], [23]. In recent years, deep learning has been widely applied in various HSI processing tasks [24], [25], [26], and has also been successfully applied to BS [27], [28], [29], [30], in which deep learning models can learn the inherent correlations and patterns between bands, and identify the most important ones.

Recently, sparse representation has found wide applications in HSI processing, such as sparse self-representation (SR) [31], sparse subspace clustering [32], sparse nonnegative factorization [33], low-rank sparse representation [34], which involves addressing an optimization problem with constraints on sparsity and has been successfully applied in BS. Li and Qi [35] utilized KSVD to decompose HSI data into the product of an overcomplete dictionary and a coefficient matrix, where sparsity of the coefficient matrix is employed to express the importance of each band. Sun et al. [36] introduced dissimilarity-weighted sparse SR, which incorporates a dissimilarity-based coding cost and sparsity constraint to guarantee that all bands can be represented by a few bands with minimal cost. Li et al. [37] designed a multidictionary sparse representation and analyzed sparse solutions for each band vector and its dictionary to determine the contribution of each band to the raw image. Wei et al. [38] presented scalable one-pass SR learning, which combines scalable SR learning with adaptive linear combinations, graph-based regularization, and a memory function in a cache. Spatial or spectral relationship that captures the structure information among pixels or bands which are important for HSI analysis [39], are gradually integrated into SR-based BS. For example, Cai et al. [40] leveraged structural information among bands by incorporating graph convolution into SR. Ma et al. [41] proposed correlation-based SR, where the band correlation is used to assign weights for representation and enforce the diversity of selected bands. Li et al. [42] constructed an initial graph for bands based on Euclidean distance, then approximated the initial affinity matrix to derive a block diagonal similarity matrix corresponding to ordered clusters, and finally selected a band from each cluster according to the band importance estimated by dictionary learning. Zhang et al. [43] incorporated spatial information from various homogeneous regions into SR by capturing relationships within the same segment through a structural graph. Zhang et al. [44] applied sparse principal component analysis with an orthogonal constraint and $\ell_{2,1}$ norm-based sparse regularization to learn the band projection weights and simultaneously preserved the local neighbor structure between pixels by leveraging the advantages of multiple predefined graphs. Wang et al. [45] proposed graph regularized spatial–spectral subspace clustering (GRSC). It first segments the HSI into homogeneous regions, and then extracts latent features through Laplacian Eigenmaps for each region to obtain a new representation of HSI. Based on the new band features, GRSC constructs subspace clustering, where a similarity graph between latent features is exploited to preserve the spatial structure in the new representation space. Tang et al. [46] proposed spatial and spectral structure preserved SR (S4P). It segments the HSI into different superpixels to get a superpixel-level feature vector for each band, then employs SR to learn the spectral relationship among bands, which involves the learning of a unified similarity across different superpixels to preserve the spatial structure through adaptive fusion of multiple predefined graphs. Zhang et al. [47] proposed robust dual graph SR (RDGSR). It constructs spatial and spectral graphs, then utilizes spatial and spectral relationships for data propagation across spatial and spectral dimensions, and conducts SR on the new hyperspectral data.

From the above analysis, it can be found that [40] employs only spectral relationship graphs for BS, while the spatial relationship is ignored. Both GRSC [45] and S4P [46] excel in meticulously designed spatial relationship graph learning, while they solely rely on SR in exploiting spectral relationships in the sparse framework. In addition, GRSC and S4P conduct all selection modeling in the processed feature space via Laplacian Eigenmaps or averaged superpixel, which may miss the influence of some important raw information. Although RDGSR [47] exploits both spatial and spectral relationships, it only integrates these relationships at the HSI data representation level, rather than at the SR level used for final BS. In addition, it constructs spatial relationships for all pixels in segmented regions, which results in extremely high computational time and space complexity, especially for large-scale high-resolution HSIs. For example, the Pavia

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University dataset has 207 400 pixels, and constructing pixel-level spatial relationships encounters out-of-memory issues on a platform with i5-13400F and 16 GB RAM. Except for the computation cost, this fine-grained spatial relationship may not be necessary for BS, and larger-scale structures within the image might be of greater interest.

Thus, this article proposes joint spatial and spectral graph-based consistent SR (JSSGCSR) shown in Fig. 1 to more effectively exploit spatial and spectral relationships for BS, which constructs exclusive spectral graph SR and spatial graph SR, respectively, and then enforces consistent sparse patterns on the joint spatial–spectral SR combination matrix to achieve a unified and robust representative band set. Specifically, instead of simply performing composite spatial and spectral data propagation on the HSI data, the proposed method separately conducts SR to handle each view of the spatial and spectral graph to better consider two different structure characteristics and reduce mutual interference and ultimately integrates two SR results to improve the BS performance by imposing consistent sparsity pattern on their joint representation coefficients. Furthermore, the two views of the graph are integrated into different data spaces with SR. Considering the importance of spectral relationships in BS and the low computational cost, spectral graph convolutional SR is performed in the original data space. However, due to the high computational complexity of spatial relationship modeling caused by the number of pixels far exceeding that of bands, spatial graph convolutional SR is conducted in the data space with greatly reduced spatial pixels, which is obtained through superpixel segmentation and pooling.

Therefore, the advantages and differences with other methods are as follows. 1) The proposed JSSGCSR explicitly captures both spatial and spectral relationships through graph representation, and two types of relationships are processed separately and integrated at the selection-making level. 2) Two different data spaces are utilized in JSSGCSR, including the original data space and the superpixel space. This not only reduces the computational cost associated with spatial relationship modeling, whose complexity is proportional to the square of spatial pixels but also avoids ignoring the original data information. Finally, the main contributions of this article are summarized as follows.

1) The JSSGCSR is proposed for BS to effectively exploit both spatial and spectral structure information in HSIs. It adopts a "separate and integrate" approach, which conducts SR separately for each view of spatial and spectral data and relationship, and merges the results of these two SRs finally with the consistent sparse constrain on joint representation coefficients.
2) In contrast to existing methods that solely model SR either in the original space or in the space after dimensionality reduction, JSSGCSR integrates spatial relationship graphs and spectral relationship graphs into different feature spaces. Spectral relationships with lower computational complexity are integrated into the original space, while spatial relationships with higher computational complexity are exploited in the feature space after spatial segmentation and pooling, which significantly reduces the spatial elements. This processing ensures the accuracy of spectral relationships while alleviating the computational burden of spatial relationships.
3) To optimize the proposed JSSGCSR model, the objective function is decomposed into two subproblems, and an efficient iterative algorithm with convergence is designed to obtain jointly consistent SR coefficients used for BS. In addition, the uniform ranking strategy is designed to process the coefficients to achieve final BS results.
4) Extensive comparison and ablation experiments on three benchmark datasets are conducted, which have demonstrated the effectiveness of the proposed method.

Below, Section II elaborates the proposed model, Section III presents optimization and its analysis, Section IV provides experimental results and discussions, while Section V gives the conclusion and analyzes the limitations.

II. Proposed Method

This section comprehensively presents the details of the proposed method and covers the following key components: spatial segmentation and pooling, spectral and spatial graph construction, the modeling of JSSGCSR, and the BS strategy.

Given an HSI $\mathcal{X} \in \mathbb{R}^{W\times H\times B}$ which captures $W \times H$ pixels and consists of B spectral bands, BS aims to select a representative subset from B bands. In the task of BS, a 3-D HSI is typically mathematically transformed into $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_B] \in \mathbb{R}^{WH\times B}$, where $\mathbf{x}_i \in \mathbb{R}^{WH}$ represents the ith band vector.

A. Spatial Segmentation and Pooling

In BS, modeling spatial relationships plays a significant role in understanding the spatial structure of HSIs. However, constructing pixel-level spatial relationships is highly complex and computationally expensive, especially for large-scale high-resolution images. Moreover, such fine-grained spatial relationships may be unnecessary, as BS could be more interested in the larger-scale structures within the image. Therefore, this article constructs the spatial relationship at the superpixel level by segmenting and aggregating images into regions with larger spatial extents, to reduce the complexity of modeling spatial relationships while preserving important spatial structural information within the image.

In this article, the entropy rate segmentation (ERS) [48] is used to achieve the segmentation outcome, which captures the spatial arrangement of various objects based on the first principal component obtained through PCA. Subsequently, the segmented maps are extrapolated to all bands, resulting in

$$\mathcal{X} = \{\mathcal{X}_1, \mathcal{X}_2, \ldots, \mathcal{X}_N\} \quad \text{s.t. } \mathcal{X}_i \cap \mathcal{X}_j = \emptyset \quad \forall 1 \le i,\ j \le N,\ i \ne j \tag{1}$$

where N is the number of superpixels, and $\mathcal{X}_i$ denotes the ith superpixel with M_i pixels and B bands, and $\sum_{i=1}^{N} M_i = WH$.

Then, $\mathcal{X}_i$ can be represented in a matrix form according to the order of the bands as follows:

$$\mathbf{X}_i = \left[\mathbf{X}_i^1, \mathbf{X}_i^2, \ldots, \mathbf{X}_i^B\right] \in \mathbb{R}^{M_i\times B} \tag{2}$$

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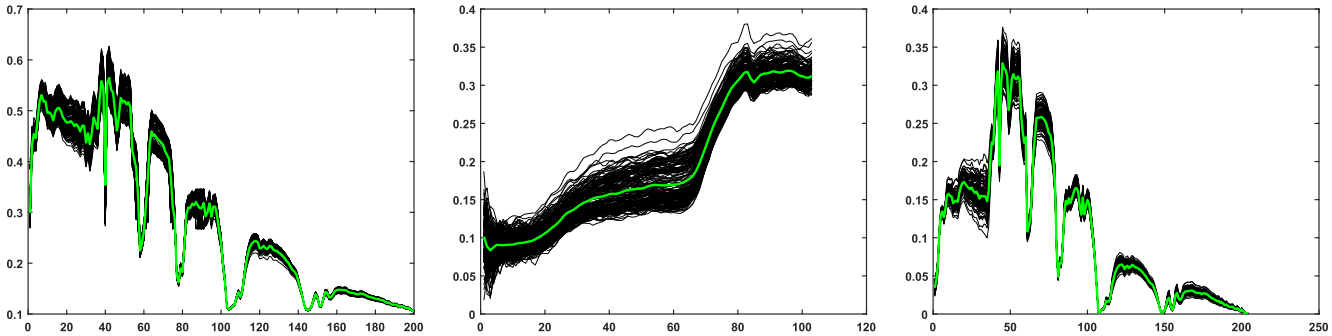


Fig. 2. Spectral curves of all single pixels (black) within a superpixel and the corresponding pooled superpixel (green) on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

where $\mathbf{X}_i^j \in \mathbb{R}^{M_i}$ is the vector formed by stretching the jth band of the ith superpixel.

Typically, neighboring pixels with a high probability of belonging to the same land cover category tend to exhibit similar spectral characteristics, particularly when they are within the same homogeneous region. Therefore, this article conducts spatial pooling on each superpixel to efficiently represent the entire superpixel, which can greatly reduce computational complexity. The spatial pooling on each superpixel can be expressed as

$$\mathbf{y}_i = \text{pool}(\mathbf{X}_i) = \left[\text{pool}\left(\mathbf{X}_i^1\right), \text{pool}\left(\mathbf{X}_i^2\right), \ldots, \text{pool}\left(\mathbf{X}_i^B\right)\right] \quad (3)$$

in which $\mathbf{y}_i \in \mathbb{R}^{1\times B}$ is the pooled result of $\mathbf{X}_i$, and pool$(\cdot)$ denotes the pooling function, which can be the average, maximum, or other statistical measures. The examples of pixels within a superpixel and the pooled superpixel are shown in Fig. 2, from which it can be observed that the pixels within a segmented superpixel have similar spectral features, and the pooled superpixel can represent the original pixels to a certain extent.

Thus, when spatial pooling is applied to all superpixels, the pooled HSI can be denoted as $\mathbf{Y} = [\mathbf{y}_1^{\mathrm{T}}, \mathbf{y}_2^{\mathrm{T}}, \ldots, \mathbf{y}_N^{\mathrm{T}}]^{\mathrm{T}} \in \mathbb{R}^{N\times B}$.

B. Spectral and Spatial Graph Construction

1) Spectral Graph: To depict the structural relationships among spectral bands with a graph $\mathcal{G}_{\text{spe}} = (\mathcal{V}_{\text{spe}}, \mathcal{E}_{\text{spe}})$, each band is considered as a vertex $v_i \in \mathcal{V}_{\text{spe}}$. To capture the relationships between these bands, the connections between neighboring spectral bands are quantified by a weighted edge $e_{ij} \in \mathcal{E}_{\text{spe}}$, which represents the strength of the connection between two spectral bands. Generally, this weighting can be determined based on various factors, such as the similarity or correlation between the band spectral characteristics.

Assume that the spectral graph is represented by the adjacency matrix $\mathbf{A}_1 \in \mathbb{R}^{B\times B}$, and $(\mathbf{A}_1)_{ij}$ denotes the edge weight between the ith and jth spectral bands. We determine this weight by considering the similarity of the bands within their local neighborhoods. Specifically, for each band $\mathbf{x}_i \in \mathbb{R}^{WH\times 1}$, its k-nearest neighbors (KNNs) are identified, denoted as $\mathcal{N}_k(\mathbf{x}_i)$. The similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$ is then quantified using the Gaussian kernel function if $\mathbf{x}_j$ belongs to $\mathcal{N}_k(\mathbf{x}_i)$. Conversely, if the bands are not within the KNNs of each other, their similarity is set to zero, indicating no direct connection in the graph. Mathematically, this can be expressed as follows:

$$(\mathbf{A}_1)_{ij} = \begin{cases} \exp\left(-\dfrac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{2\sigma^2}\right), & \mathbf{x}_i \in \mathcal{N}_k(\mathbf{x}_j) \\ & \text{or } \mathbf{x}_j \in \mathcal{N}_k(\mathbf{x}_i) \\ 0, & \text{otherwise} \end{cases} \quad (4)$$

where σ is a parameter of the Gaussian kernel controlling the width of the similarity function. This graph-based representation provides a powerful tool for exploring and understanding the complex relationships that exist within spectral data.

2) Spatial Graph: The above approach can also be employed to construct spatial graph $\mathcal{G}_{\text{spa}} = (\mathcal{V}_{\text{spa}}, \mathcal{E}_{\text{spa}})$. In this context, the only difference lies in the need to replace each band with each pooled superpixel. Specifically, each superpixel is defined as a vertex, and edge weights are assigned based on the similarity between superpixels. Analogous to the case of spectral bands, if two superpixels are close, the edge weight between them will be higher; conversely, if they are far apart, the weight will be lower or even zero. The spatial graph can also be represented by the corresponding adjacency matrix $\mathbf{A}_2 \in \mathbb{R}^{N\times N}$ as follows:

$$(\mathbf{A}_2)_{ij} = \begin{cases} \exp\left(-\dfrac{\|\mathbf{y}_i - \mathbf{y}_j\|^2}{2\sigma^2}\right), & \mathbf{y}_i \in \mathcal{N}_k(\mathbf{y}_j) \\ & \text{or } \mathbf{y}_j \in \mathcal{N}_k(\mathbf{y}_i) \\ 0, & \text{otherwise.} \end{cases} \quad (5)$$

Note that the construction of the spatial adjacency matrix $\mathbf{A}_2$ in this article is computationally efficient because we use superpixels instead of individual pixels. By grouping similar pixels into superpixels, the number of entities to consider is significantly reduced, which in turn reduces the computational complexity. This is particularly useful for large images where processing at the pixel level would be prohibitively expensive.

C. JSSGCSR Modeling

Given the HSI denoted as $\mathbf{X} \in \mathbb{R}^{WH\times B}$, SR-based BS commonly involves the introduction of a representation error term and a sparsity term, which can be generally formulated as

$$\min_{\mathbf{W}} \left\|\mathbf{X} - \mathbf{X}\mathbf{W}\right\|_F^2 + \beta\mathcal{R}(\mathbf{W}) \quad (6)$$

where the first term quantifies the error between the original HSI and its representation, the second term imposes sparsity

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constraints on the coefficient matrix $\mathbf{W} \in \mathbb{R}^{B\times B}$, and $\beta > 0$ serves as a tradeoff parameter between the two terms.

In typical practices, sparse regularization commonly uses the Frobenius norm and the $\ell_{2,1}$ norm for matrix $\mathbf{W}$. In this article, we use the $\ell_{2,1}$ norm because of its robustness, which is calculated as follows:

$$\|\mathbf{W}\|_{2,1} = \sum_{i=1}^{n} \sqrt{\sum_{j=1}^{n} \mathbf{W}_{ij}^2} = \sum_{i=1}^{n} \|\mathbf{W}_i\| \tag{7}$$

where $\mathbf{W}_i$ is the ith row of $\mathbf{W}$. Accordingly, the model encourages $\mathbf{W}$ to have only a limited number of nonzero rows.

However, the basic model in (6) only formulates the reconstruction of spectral bands, while overlooking crucial structural information and relationships, which limits the performance of BS. As analyzed in Section II-B, the spectral graph depicts the spectral relationships between different bands, while the spatial graph captures the relationships between pair-wise superpixels. It is indeed beneficial to explore and exploit the two kinds of graphs to understand the spatial and spectral characteristics of HSI data, which enables more effective BS. Therefore, this article simultaneously integrates two graphs, representing both spatial and spectral relationships, into the SR-based BS framework in (6).

First, the spectral graph is introduced into the original HSI representation, by using the graph convolution operation from graph convolutional networks (GCNs) [49] as follows:

$$\min_{\mathbf{W}_1} \|\mathbf{X} - \mathbf{X}\mathbf{A}_1\mathbf{W}_1\|_F^2 + \beta\|\mathbf{W}_1\|_{2,1} \tag{8}$$

where the graph convolution is achieved through the multiplication of the original HSI matrix $\mathbf{X}$ and the spectral adjacency matrix $\mathbf{A}_1$. As a result, each column of $\mathbf{X}\mathbf{A}_1$ not only includes its own spectral information but also incorporates the contributions of its neighboring bands.

Second, the spatial graph is introduced into the pooled HSI representation as follows:

$$\min_{\mathbf{W}_2} \|\mathbf{Y} - \mathbf{A}_2\mathbf{Y}\mathbf{W}_2\|_F^2 + \beta\|\mathbf{W}_2\|_{2,1} \tag{9}$$

where the graph convolution is achieved through the multiplication of the spatial adjacency matrix $\mathbf{A}_2$ and the pooled HSI matrix $\mathbf{X}$. As a result, by utilizing this spatial graph convolution, each row of $\mathbf{A}_2\mathbf{Y}$ not only encompasses the information of a single superpixel but also captures the information of the neighboring superpixels in space.

Therefore, to simultaneously leverage both types of structured information, a joint spatial and spectral graph-based SR model is established by combining (8) and (9) as follows:

$$\min_{\mathbf{W}_1,\mathbf{W}_2} \|\mathbf{X} - \mathbf{X}\mathbf{A}_1\mathbf{W}_1\|_F^2 + \alpha\|\mathbf{Y} - \mathbf{A}_2\mathbf{Y}\mathbf{W}_2\|_F^2 + \beta\left(\|\mathbf{W}_1\|_{2,1} + \|\mathbf{W}_2\|_{2,1}\right) \tag{10}$$

where α and β are tradeoff parameters. This model establishes respective SR for each type of graph, which provides flexibility in adaption to different graphs, and allows better utilization of the information they contain.

However, the joint model in (10) only independently models two types of graph representation, neither achieves their interaction and fusion nor addresses potential inconsistencies in the sparsity patterns of the two representations, which may lead to inconsistent BS between $\mathbf{W}_1$ and $\mathbf{W}_2$. To address this issue, we should introduce interaction and fusion mechanisms into the model. Therefore, this article employs a joint sparse regularization term that simultaneously constrains the sparsity of both graph representations into the optimization objective to maintain consistent sparsity and ensure consistent BS.

As a result, the loss function or the objective function of the proposed JSSGCSR for HSI BS can be finally formulated as follows:

$$\begin{aligned}\min_{\mathbf{W}_1,\mathbf{W}_2} &\|\mathbf{X} - \mathbf{X}\mathbf{A}_1\mathbf{W}_1\|_F^2 + \alpha\|\mathbf{Y} - \mathbf{A}_2\mathbf{Y}\mathbf{W}_2\|_F^2 \\ &+ \beta\left(\|\mathbf{W}_1\|_{2,1} + \|\mathbf{W}_2\|_{2,1}\right) + \gamma\|\mathbf{W}_c\|_{2,1} \\ &\text{s.t. } \mathbf{W}_c = [\mathbf{W}_1, \mathbf{W}_2], \quad \mathbf{W}_c \in \mathbb{R}^{B\times 2B}\end{aligned} \tag{11}$$

where $\mathbf{W}_c = [\mathbf{W}_1, \mathbf{W}_2] \in \mathbb{R}^{B\times 2B}$ is the spatial and spectral joint graph SR matrix. Our objective function is designed as a comprehensive loss function, incorporating the following components: spectral graph SR error, spatial graph SR error, individual sparsity constraints for each SR, and a joint consistent sparsity constraint. Through this improvement, the model can better leverage information from both graphs, maintain consistency in sparsity patterns, and enhance the robustness and consistency of BS.

Overall, our proposed method offers a principled way to integrate spatial and spectral information in HSIs, by leveraging the power of graph convolution within the SR framework for robust and effective BS.

D. BS Strategy

After solving the JSSGCSR model which will be detailed in Section III, the SR coefficients can be obtained. Then, how to select bands becomes the final crucial step. Currently, there are two methods: one is the ranking [36], [37], [38], [40], which computes the ℓ_1 or ℓ_2 norm of the coefficient vector for each band and sorts them in descending order; the other is spectral clustering [40], [45], [50], [51], which constructs an adjacency matrix based on the coefficient matrix and then performs spectral clustering to select informative bands. The ranking method is efficient and easy to implement, while highly similar bands could be simultaneously selected due to their similar coefficients. On the other hand, spectral clustering can extract more representative combinations of bands, but it comes with higher computational complexity. Moreover, when the number of selected bands changes, reclustering is required. In addition, spectral clustering typically requires a square matrix as input, which is often referred to as the similarity matrix or adjacency matrix. This limits its scope of application to some extent, especially when dealing with the joint of multiple SR coefficient matrices, such as $\mathbf{W}_c \in \mathbb{R}^{B\times 2B}$ in this article.

Therefore, this article proposes the "uniform ranking." First, it partitions the HSI Bands into K equal-sized subspaces. It can be understood as grouping adjacent bands into clusters with equal band numbers. Then, it selects the most representative band within each subspace. Since the ℓ_2 norm can be used as a measure of the importance or saliency of bands, the band with the highest ℓ_2 norm within each subspace is then

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selected as the most representative band for that subspace. This approach is simple and intuitive while considering both the high correlation of adjacent bands and the importance of SR, which can achieve a more balanced and effective BS outcome.

III. Optimization of JSSGCSR

This section introduces the algorithm designed to solve the JSSGCSR model and provides a theoretical analysis of its convergence and computational complexity.

A. Optimization Design

Due to the involving of the $\ell_{2,1}$ norm, the problem in (11) becomes nonsmooth, making direct optimization challenging. Inspired by the optimization method in [52], we propose an iterative algorithm to efficiently solve (11) alternatively.

According to the half-quadratic theory, the problem in (11) can be converted the following problem:

$$\begin{aligned}\min_{\mathbf{W}_1,\mathbf{W}_2} &\left\|\mathbf{X}-\mathbf{X}\mathbf{A}_1\mathbf{W}_1\right\|_F^2+\alpha\left\|\mathbf{Y}-\mathbf{A}_2\mathbf{Y}\mathbf{W}_2\right\|_F^2\\&+\beta\left(\mathrm{Tr}\left(\mathbf{W}_1^{\mathrm{T}}\mathbf{D}_1\mathbf{W}_1\right)+\mathrm{Tr}\left(\mathbf{W}_2^{\mathrm{T}}\mathbf{D}_2\mathbf{W}_2\right)\right)\\&+\gamma\mathrm{Tr}\left(\mathbf{W}_c^{\mathrm{T}}\mathbf{D}_c\mathbf{W}_c\right)\end{aligned}\tag{12}$$

where $\mathbf{D}_1, \mathbf{D}_2$, and $\mathbf{D}_c \in \mathbb{R}^{B\times B}$ are all diagonal matrices, and their ith diagonal element is defined as

$$(\mathbf{D}_1)_{ii}=\frac{1}{2\sqrt{\|(\mathbf{W}_1)_i\|^2+\epsilon}}\tag{13}$$

$$(\mathbf{D}_2)_{ii}=\frac{1}{2\sqrt{\|(\mathbf{W}_2)_i\|^2+\epsilon}}\tag{14}$$

$$(\mathbf{D}_c)_{ii}=\frac{1}{2\sqrt{\|(\mathbf{W}_c)_i\|^2+\epsilon}}.\tag{15}$$

where ϵ serves as a smoothing parameter and is typically set to a small constant value.

Optimizing (12) with respect to $\mathbf{W}_1$ and $\mathbf{W}_2$ is tantamount to optimizing the following two subproblems:

$$\min_{\mathbf{W}_1}\left\|\mathbf{X}-\mathbf{X}\mathbf{A}_1\mathbf{W}_1\right\|_F^2+\beta\mathrm{Tr}\left(\mathbf{W}_1^{\mathrm{T}}\mathbf{D}_1\mathbf{W}_1\right)+\gamma\mathrm{Tr}\left(\mathbf{W}_1^{\mathrm{T}}\mathbf{D}_c\mathbf{W}_1\right)\tag{16}$$

$$\min_{\mathbf{W}_2}\alpha\left\|\mathbf{Y}-\mathbf{A}_2\mathbf{Y}\mathbf{W}_2\right\|_F^2+\beta\mathrm{Tr}\left(\mathbf{W}_2^{\mathrm{T}}\mathbf{D}_2\mathbf{W}_2\right)+\gamma\mathrm{Tr}\left(\mathbf{W}_2^{\mathrm{T}}\mathbf{D}_c\mathbf{W}_2\right).\tag{17}$$

With $\mathbf{D}_1, \mathbf{D}_2$, and $\mathbf{D}_c$ fixed, the optimal solution of (16) and (17) can be computed by setting the derivatives of the objective functions with respect to $\mathbf{W}_1$ and $\mathbf{W}_2$ to zero, yielding the following equations:

$$\left(\mathbf{A}_1^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{A}_1+\beta\mathbf{D}_1+\gamma\mathbf{D}_c\right)\mathbf{W}_1=\mathbf{A}_1^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\tag{18}$$

$$\left(\alpha\mathbf{Y}^{\mathrm{T}}\mathbf{A}_2^{\mathrm{T}}\mathbf{A}_2\mathbf{Y}+\beta\mathbf{D}_2+\gamma\mathbf{D}_c\right)\mathbf{W}_2=\alpha\mathbf{Y}^{\mathrm{T}}\mathbf{A}_2^{\mathrm{T}}\mathbf{Y}.\tag{19}$$

Therefore, $\mathbf{W}_1$ and $\mathbf{W}_2$ can be computed by

$$\mathbf{W}_1=\left(\mathbf{A}_1^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\mathbf{A}_1+\beta\mathbf{D}_1+\gamma\mathbf{D}_c\right)^{-1}\left(\mathbf{A}_1^{\mathrm{T}}\mathbf{X}^{\mathrm{T}}\mathbf{X}\right)\tag{20}$$

$$\mathbf{W}_2=\left(\alpha\mathbf{Y}^{\mathrm{T}}\mathbf{A}_2^{\mathrm{T}}\mathbf{A}_2\mathbf{Y}+\beta\mathbf{D}_2+\gamma\mathbf{D}_c\right)^{-1}\left(\alpha\mathbf{Y}^{\mathrm{T}}\mathbf{A}_2^{\mathrm{T}}\mathbf{Y}\right).\tag{21}$$

By iteratively updating $\mathbf{D}_1, \mathbf{D}_2$, and $\mathbf{D}_c$ and solving $\mathbf{W}_1, \mathbf{W}_2$, and $\mathbf{W}_c = [\mathbf{W}_1, \mathbf{W}_2]$ that minimizes the objective function in (11) can be obtained. Then, the informative bands can be selected based on the representation coefficient matrix $\mathbf{W}_c$.

Accordingly, the whole process of the proposed JSSGCSR method for BS is summarized in Algorithm 1.

Algorithm 1 BS Process of JSSGCSR

Input: a hyperspectral image $\mathbf{X} \in \mathbb{R}^{WH\times B}$, parameters α, β and γ, and maximal iterations $T_{\max}$ (default to 50).
Output: K selected bands.

1: *// Step 1: Spatial Segmentation and Pooling //*
2: Obtaining $\mathbf{Y} \in \mathbb{R}^{N\times B}$ by using entropy rate segmentation and conducting average or max pooling on each segmented superpixel according to (3).
3: *// Step 2: Spectral and Spatial Graph Construction //*
4: Obtaining spectral graph adjacency matrix $\mathbf{A}_1$ by (4).
5: Obtaining spatial graph adjacency matrix $\mathbf{A}_2$ by (5).
6: *// Step 3: Solving the Objective Function in (11) //*
7: *Initialization*: $t = 0$, randomizing $\mathbf{W}_1, \mathbf{W}_2$.
8: **while** $t \le T_{\max}$ **do**
9: Computing $\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_c$ through (13), (14), and (15).
10: Computing $\mathbf{W}_1, \mathbf{W}_2$ through (20) and (21).
11: Updating $\mathbf{W}_c$ as $\mathbf{W}_c = [\mathbf{W}_1, \mathbf{W}_2]$.
12: Increasing $t \leftarrow t + 1$.
13: **end while**
14: *// Step 4: Band Selection Based on $\mathbf{W}_c$ //*
15: Calculating the ℓ_2 norm of each row of $\mathbf{W}_c$ to form a curve, and dividing the curve evenly into K segments.
16: Selecting the band corresponding to the maximum value within each segment of the ℓ_2 norm curve.

B. Convergence Proof

In this section, we discuss the convergence of Algorithm 1. The value of the objective function (11) will monotonically decrease through iterations.

Let $\mathcal{J}_1(\mathbf{W}_1, \mathbf{W}_2) = \|\mathbf{X}-\mathbf{X}\mathbf{A}_1\mathbf{W}_1\|_F^2 + \alpha\|\mathbf{Y}-\mathbf{A}_2\mathbf{Y}\mathbf{W}_2\|_F^2$. After the tth iteration, $\mathbf{W}_1^t, \mathbf{W}_2^t, \mathbf{D}_1^t, \mathbf{D}_2^t$, and $\mathbf{D}_c^t$ is known. By using (20) and (21), the optimal solution $\mathbf{W}_1^{t+1}, \mathbf{W}_2^{t+1}$ can be calculated. Therefore, the following inequality holds:

$$\begin{aligned}&\beta\left(\mathrm{Tr}\left(\left(\mathbf{W}_1^{t+1}\right)^{\mathrm{T}}\mathbf{D}_1^t\mathbf{W}_1^{t+1}\right)+\mathrm{Tr}\left(\left(\mathbf{W}_2^{t+1}\right)^{\mathrm{T}}\mathbf{D}_2^t\mathbf{W}_2^{t+1}\right)\right)\\&\quad+\gamma\mathrm{Tr}\left(\left(\mathbf{W}_c^{t+1}\right)^{\mathrm{T}}\mathbf{D}_c^t\mathbf{W}_c^{t+1}\right)+\mathcal{J}_1\left(\mathbf{W}_1^{t+1},\mathbf{W}_2^{t+1}\right)\\&\le\beta\left(\mathrm{Tr}\left(\left(\mathbf{W}_1^{t}\right)^{\mathrm{T}}\mathbf{D}_1^t\mathbf{W}_1^{t}\right)+\mathrm{Tr}\left(\left(\mathbf{W}_2^{t}\right)^{\mathrm{T}}\mathbf{D}_2^t\mathbf{W}_2^{t}\right)\right)\\&\quad+\gamma\mathrm{Tr}\left(\left(\mathbf{W}_c^{t}\right)^{\mathrm{T}}\mathbf{D}_c^t\mathbf{W}_c^{t}\right)+\mathcal{J}_1\left(\mathbf{W}_1^{t},\mathbf{W}_2^{t}\right).\end{aligned}\tag{22}$$

Substituting (13)–(15) to the inequality (22), we have the following:

$$\begin{aligned}&\beta\sum_{i=1}^{B}\left(\frac{\left\|\left(\mathbf{W}_1^{t+1}\right)_i\right\|^2+\epsilon}{2\sqrt{\left\|\left(\mathbf{W}_1^{t}\right)_i\right\|^2+\epsilon}}+\frac{\left\|\left(\mathbf{W}_2^{t+1}\right)_i\right\|^2+\epsilon}{2\sqrt{\left\|\left(\mathbf{W}_2^{t}\right)_i\right\|^2+\epsilon}}\right)\\&+\gamma\sum_{i=1}^{B}\frac{\left\|\left(\mathbf{W}_c^{t+1}\right)_i\right\|^2+\epsilon}{2\sqrt{\left\|\left(\mathbf{W}_c^{t}\right)_i\right\|^2+\epsilon}}+\mathcal{J}_1\left(\mathbf{W}_1^{t+1},\mathbf{W}_2^{t+1}\right)\end{aligned}$$

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TABLE I
COMPUTATIONAL COMPLEXITY OF THE OPTIMIZATION OF JSSGCSR

Computation	Complexity
updating $\mathbf{D}_1, \mathbf{D}_2, \mathbf{D}_c$	$\mathcal{O}(B^2)$
updating $\mathbf{W}_1$	$\max\{\mathcal{O}(WHB^2), \mathcal{O}(B^3)\}$
updating $\mathbf{W}_2$	$\max\{\mathcal{O}(N^2B), \mathcal{O}(B^3)\}$
Whole Algorithm	$\max\{\mathcal{O}(T_{max}WHB^2), \mathcal{O}(T_{max}BN^2)\}$

$$\leq \beta \sum_{i=1}^{B} \left(\frac{\left\| \left(\mathbf{W}_1^t\right)_i \right\|^2 + \epsilon}{2\sqrt{\left\| \left(\mathbf{W}_1^t\right)_i \right\|^2 + \epsilon}} + \frac{\left\| \left(\mathbf{W}_2^t\right)_i \right\|^2 + \epsilon}{2\sqrt{\left\| \left(\mathbf{W}_2^t\right)_i \right\|^2 + \epsilon}} \right) + \gamma \sum_{i=1}^{B} \frac{\left\| \left(\mathbf{W}_c^t\right)_i \right\|^2 + \epsilon}{2\sqrt{\left\| \left(\mathbf{W}_c^t\right)_i \right\|^2 + \epsilon}} + \mathcal{J}_1\left(\mathbf{W}_1^t, \mathbf{W}_2^t\right). \tag{23}$$

According to the Lemma 1 in [52], the following inequality about $\mathbf{W}_1$ holds:

$$\sum_{i=1}^{B} \sqrt{\left\| \left(\mathbf{W}_1^{t+1}\right)_i \right\|^2 + \epsilon} - \sum_{i=1}^{B} \frac{\left\| \left(\mathbf{W}_1^{t+1}\right)_i \right\|^2 + \epsilon}{2\sqrt{\left\| \left(\mathbf{W}_1^t\right)_{i\cdot} \right\|^2 + \epsilon}} \leq \sum_{i=1}^{B} \sqrt{\left\| \left(\mathbf{W}_1^t\right)_i \right\|^2 + \epsilon} - \sum_{i=1}^{B} \frac{\left\| \left(\mathbf{W}_1^t\right)_i \right\|^2 + \epsilon}{2\sqrt{\left\| \left(\mathbf{W}_1^t\right)_{i\cdot} \right\|^2 + \epsilon}}. \tag{24}$$

The similar inequality also holds for $\mathbf{W}_2$ and $\mathbf{W}_c$.

Based on (23) and (24), we have

$$\begin{aligned} &\beta \sum_{i=1}^{B} \left(\sqrt{\left\| \left(\mathbf{W}_1^{t+1}\right)_i \right\|^2 + \epsilon} + \sqrt{\left\| \left(\mathbf{W}_2^{t+1}\right)_i \right\|^2 + \epsilon} \right) + \gamma \sum_{i=1}^{B} \sqrt{\left\| \left(\mathbf{W}_c^{t+1}\right)_i \right\|^2 + \epsilon} + \mathcal{J}_1\left(\mathbf{W}_1^{t+1}, \mathbf{W}_2^{t+1}\right) \\ &\leq \beta \sum_{i=1}^{B} \left(\sqrt{\left\| \left(\mathbf{W}_1^t\right)_i \right\|^2 + \epsilon} + \sqrt{\left\| \left(\mathbf{W}_2^t\right)_i \right\|^2 + \epsilon} \right) + \gamma \sum_{i=1}^{B} \sqrt{\left\| \left(\mathbf{W}_c^t\right)_i \right\|^2 + \epsilon} + \mathcal{J}_1\left(\mathbf{W}_1^t, \mathbf{W}_2^t\right). \end{aligned} \tag{25}$$

Thus, it is demonstrated that the objective function in (11) monotonically decreases through iteration.

C. Computational Complexity

The computational complexity of JSSGCSR's optimization is detailed in Table I. Given that the number of pixels is usually greater than that of spectral bands, that is, $WH \gg B$, the computational complexity of the designed optimization can be roughly estimated as $\max\{\mathcal{O}(T_{\max} WHB^2), \mathcal{O}(T_{\max} BN^2)\}$.

In addition, the computational complexity of preprocessing before model optimization, including spatial segmentation and pooling, spatial and spectral relationship construction, is about $\mathcal{O}(B^2WH + WH \log WH + N^2B + N^2 \log N + B^2 \log B)$, which can be approximated as $\mathcal{O}(B^2WH + N^2B)$ due to $WH \gg B, N$.

IV. EXPERIMENTAL RESULTS

This section presents the experimental results and discussions, which mainly include experiment setup, comparisons with state-of-the-art methods, computational efficiency, ablation studies, and parameter exploration. The results demonstrate the effectiveness of the proposed method, highlighting its advantages and providing insights into its performance under various conditions.

A. Experiment Setup

1) Datasets: Three benchmark datasets commonly employed in existing studies are utilized in our experiments, and their details are as follows.

a) Indian Pines: It was captured by the AVIRIS sensor over the Indian Pines test site in Northwestern Indiana. It comprises 145 × 145 pixels and offers a spatial resolution of 20 m. Featuring 224 spectral bands with a spectral resolution of 10 nm, the dataset retains 200 bands after the elimination of 20 water absorption bands [104–108], [150–163], and 220 [53]. This dataset has 16 classes, including Alfalfa, Corn-notill, Corn-mintill, Corn, Grass-pasture, Grass-trees, Grass-pasture-mowed, Hay-windrowed, Oats, Soybean-notill, Soybean-mintill, Soybean-clean, Wheat, Woods, Buildings-Grass-Trees-Drives, and Stone-Steel-Towers.

b) Pavia University: It was captured by the ROSIS sensor over Pavia University, Italy. It consists of 610 × 340 pixels with a spatial resolution of 1.3 m and encompasses 115 bands ranging from 0.43 to 0.86 μm. After the elimination of 12 noisy bands [53], 103 bands remain in this dataset. This dataset comprises nine classes, including asphalt, meadows, gravel, trees, painted metal sheets, bare soil, bitumen, self-blocking bricks, and shadows.

c) Salinas: It was collected by the AVIRIS sensor over Salinas Valley, California. It comprises 512 × 217 pixels with a spatial resolution of 3.7 m and includes 224 bands spanning wavelengths from 0.4 to 2.5 μm. After the exclusion of 20 bands due to water absorption and low SNR, 204 bands are retained in the dataset. This dataset encompasses 16 classes, including Brocoli-green-weeds-1, Brocoli-green-weeds-2, Fallow, Fallow-rough-plow, Fallow-smooth, Stubble, Celery, Grapes-untrained, Soil-vinyard-develop, Corn-senesced-green-weeds, Lettuce-romaine-4wk, Lettuce-romaine-5wk, Lettuce-romaine-6wk, Lettuce-romaine-7wk, Vinyard-untrained, and Vinyard-vertical-trellis.

In Table II, the details of Indian Pines, Pavia University, and Salinas datasets are summarized.

2) Comparison Methods: The following methods are compared to the proposed one.

1) Ranking-based methods: including MVPCA [6] and SR-SSIM [54].
2) Clustering-based methods: including E-FDPC [17].
3) Subspace partitioning-based methods: including CDSP-MinV [20] and MCBS [21].
4) SR-based methods: including SMRS [55], MDSR [37], GCSR [40], RDGSR [47], GRSC [45], and S4P [46].

All methods use their original default parameters or the best parameters obtained through searching.

3) Classification Settings: The performance of different BS approaches is evaluated through classification using the selected bands. Classification accuracy is assessed using KNNs and support vector machine (SVM). In addition, a random

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TABLE II
SUMMARY OF INDIAN PINES, PAVIA UNIVERSITY, AND SALINAS DATASETS

	Resolution	Pixels	Background	Objects	Bands	Available Bands	Classes	Sensors
Indian Pines	145×145	21025	10776	10249	224	200	16	AVIRIS
Pavia University	610×340	207400	164624	42776	115	103	9	ROSIS
Salinas	512×217	111104	56975	54129	224	204	16	AVIRIS

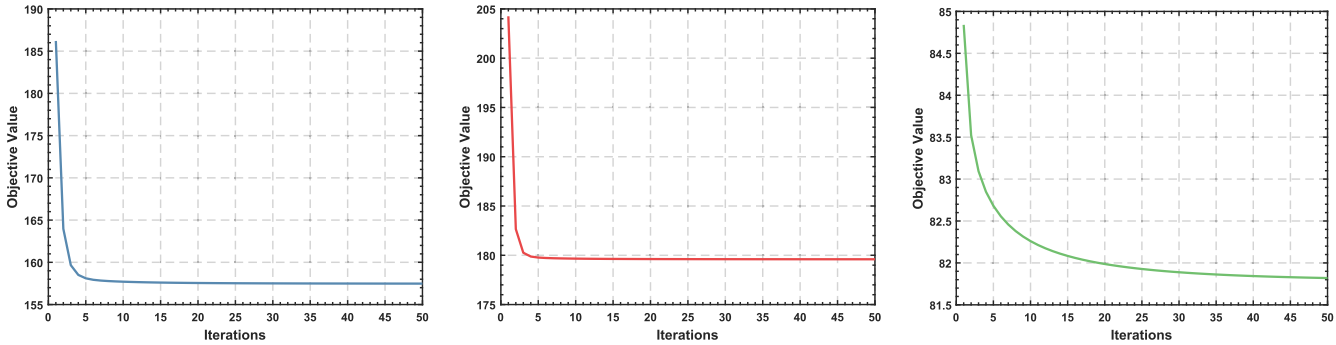


Fig. 3. Objective values with respect to the iterations numbers on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

selection of 10% of the samples from each class is utilized for training, while the remaining samples are used for testing. To mitigate the impact of training sample randomness, the reported results are obtained by averaging the outcomes of ten individual tests. The classification measures employed include overall accuracy (OA), average accuracy (AA), and Kappa.

4) Parameters Settings: For parameter settings of the proposed method, α is set as 10, γ is set as 0.001, and β is set as 0.01,0.001,10 for Indian Pines, Pavia University, and Salinas, respectively. The number of superpixels N is set as 1% of the number of original pixels, and the number of neighbors k in KNN for spatial and spectral graph construction is set as 5 for all three datasets.

B. Convergence of JSSGCSR Optimization

The convergence of the proposed algorithm has been theoretically proven in Section III-B, and visual experimental results are presented in this section. The changes of the objective function in (11) with respect to the number of iterations are shown in Fig. 3 for the three datasets. It can be observed that as the number of iterations increases, the value of the objective function gradually decreases, indicating that the algorithm is gradually approaching the optimal solution. In addition, it is found that the convergence speed of the designed algorithm is quite rapid, particularly for Indian Pines and Pavia University. Therefore, the maximum number of iterations is set to 50 in our experiments.

C. Comparisons of Classification

These experiments aim to compare the classification performance by varying the size of selected bands of different BS methods. Specifically, the size of the band subset ranges from [5:5:50]. On each dataset, two sets of results will be compared: 1) OA curves, representing OA values corresponding to the number of selected bands; and 2) the average performance across all different numbers of bands.

1) OA Under Different Number of Bands: Under different numbers of selected bands, the classification results in terms of OA for compared methods are depicted in Figs. 4 and 5, where Fig. 4 illustrates the results for the KNN classifier, while Fig. 5 presents the results for the SVM classifier. The following conclusions can be drawn from the figures.

1) The proposed method achieves the highest OA across all three datasets, regardless of whether KNN or SVM is used. This demonstrates that incorporating both spatial and spectral characteristics can enhance the performance of BS. By considering spatial context alongside spectral signatures, the proposed method can better capture the structural relationships within the HSI data, leading to more informative and representative BSs.
2) Across all the compared BS methods on each dataset with KNN and SVM classifiers, it is observed that as the number of selected bands increases, the OA gradually increases and eventually stabilizes. This trend can be attributed to the expanding richness of spectral information, which enables the classifier to more accurately distinguish between different land cover classes. However, beyond a certain point, further increasing the number of bands may not significantly enhance classification performance, and could potentially lead to higher computational cost.
3) For the same BS method, the OA under KNN and SVM exhibits similar change patterns but with SVM generally achieving higher OA values compared to KNN. The better classification performance of SVM could be attributed to its ability to learn more complex decision boundaries than KNN, which typically relies on simple majority voting within neighborhoods.
4) Across different datasets, there are discernible differences in OA. Generally, it can be observed that the OA is lower on Indian Pines compared to Pavia University and Salinas. This may be due to various factors such as differences in the complexity of the land cover classes, the resolution of the imagery, or the amount of training samples.

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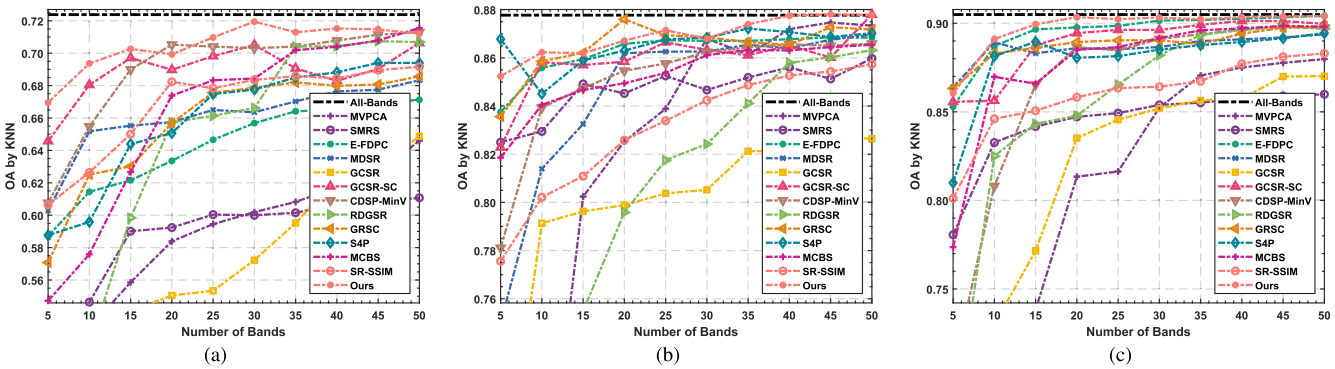


Fig. 4. Comparison of OA by KNN with respect to different band numbers on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

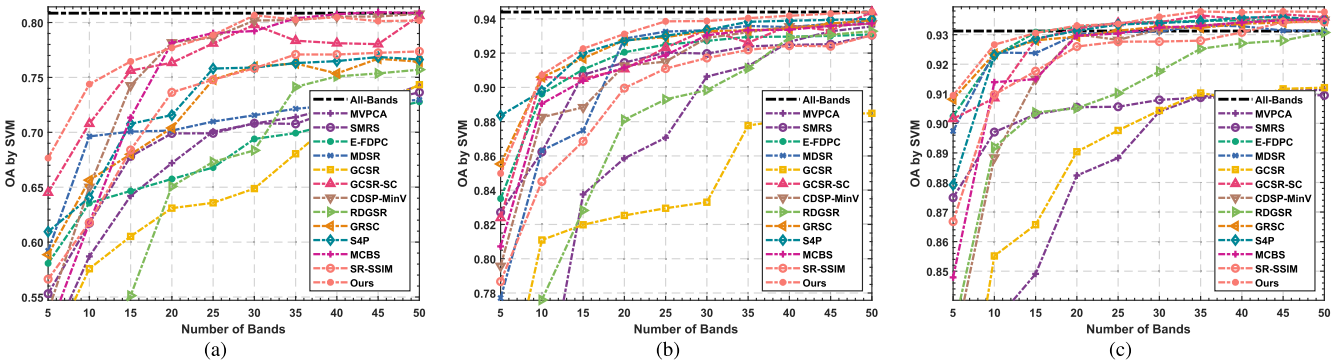


Fig. 5. Comparison of OA by SVM with respect to different band numbers on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

In addition to the above overall analysis, the specific analysis for each method is as follows.

1) For ranking-based methods, MVPCA [6] typically has lower OA among the compared methods. This is due to that MVPCA may overfocus on reducing the variance while neglecting the redundancy and complementarity among bands. In addition, MVPCA may be susceptible to noise and irrelevant information. Another ranking-based BS method, SR-SSIM [54], introduces SSIM [56] to better capture spatial–spectral information and quantify relationships between bands. Thus, its accuracy of BS and classification is enhanced.
2) The clustering-based method, E-FDPC [17], combining the benefits of ranking and clustering techniques, can achieve relatively satisfactory results among these methods. It employs fast-density peak clustering to group similar bands together and selects one or more representative bands from each cluster. Consequently, the selected bands can maximize the informational content of the entire cluster while minimizing redundancy. However, it is also noted that E-FDPC does not exhibit the high performance observed in Pavia University and Salinas on Indian Pines. This may be related to the noise that interferes with the detection of density peaks, the parameters that have not been fine-tuned, and the imbalanced class that causes poor performance in identifying minority classes.
3) Subspace portioning-based methods generally can achieve relatively good performance. Specifically, CDSP-MinV [20] divides the HSI into subspaces based on intraclass and interclass correlation distance, then measures the band priority based on the variance to select the most representative band in each subspace. MCBS [21] also conducts subspace portioning based on correlations and then selects bands from each subspace based on multiple band prioritization criteria. It can be found that this kind of BS integrates the advantages of grouping and ranking, resulting in the selected bands having low correlation and strong representation ability.
4) SR-based BS methods, particularly those enhanced ones, typically yield quite good results. As a basic model, SMRS [55] solely focuses on modeling the sparse reconstruction of HSI data, resulting in limited performance. MDSR [37] enhances the performance by constructing a sparse representation for each band with the corresponding dictionary, which allows for a more fine-grained sparse coding process to better capture complex relationships and structures within the HSI. Among these methods, GCSR [40] performs poorly mainly due to two reasons. First, the sparse constraint induced by the Frobenius norm does not promote sparsity at the individual band level, making it difficult to effectively distinguish the contributions of each band. Second, GCSR selects bands based on the ranking of representation coefficients, which may lead to highly correlated bands being selected, especially the adjacent ones. GCSR-SC [40] enhances the performance of GCSR by replacing the ranking strategy with spectral clustering, which significantly improves the classification accuracy. By applying spectral clustering to the adjacency matrix derived from the SR coefficient matrix, GRSR-SC incorporates both local and global information when selecting bands, leading to more discriminative and diverse band combinations. RDGSR [47] integrates both spatial and

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TABLE III
AVERAGE AND STANDARD DEVIATION OF EVALUATION METRICS OF KNN OVER DIFFERENT NUMBER (5:5:50) OF SELECTED BANDS

Dataset	Metrics (%)		All Bands	MVPCA	SMRS	E-FDPC	MDSR	GCSR	GCSR-SC	CDSP-MinV	RDGSR	GRSC	S4P	MCBS	SR-SSIM	Ours
Indian Pines	OA	*mean*	72.38	58.05	58.84	64.30	66.04	56.98	68.86	69.01	64.35	65.66	65.92	66.21	66.79	**70.49**
		std	0.00	6.00	2.82	2.81	2.26	5.72	1.73	3.36	7.59	3.73	3.94	5.89	2.98	1.48
	AA	*mean*	62.75	48.33	50.25	52.57	56.48	47.56	60.31	59.52	51.70	56.63	55.62	55.82	55.94	59.96
		std	0.00	6.92	2.49	2.47	2.83	5.64	1.47	3.61	7.82	3.77	3.55	6.70	3.21	1.38
	Kappa	*mean*	70.19	55.66	56.47	61.94	63.68	54.57	66.58	66.76	62.03	63.33	63.58	63.89	64.46	**68.26**
		std	0.00	6.05	2.83	2.84	2.30	5.76	1.76	3.43	7.65	3.78	3.99	6.00	3.04	1.51
Pavia University	OA	*mean*	87.77	80.58	84.67	86.25	84.50	79.64	85.91	85.04	80.38	86.47	86.53	85.26	83.04	**86.90**
		std	0.00	10.20	1.11	0.97	3.79	4.58	1.41	2.59	5.91	1.14	0.81	1.47	2.69	0.84
	AA	*mean*	85.40	75.24	82.10	83.53	81.21	76.18	83.11	81.87	75.78	83.84	83.46	82.13	79.39	**84.20**
		std	0.00	13.24	1.35	1.09	4.75	4.99	1.57	3.66	7.40	1.27	0.88	2.21	3.34	1.18
	Kappa	*mean*	84.41	76.14	80.67	82.56	80.49	74.78	82.15	81.11	75.68	82.82	82.89	81.37	78.70	**83.36**
		std	0.00	11.40	1.32	1.16	4.43	5.05	1.68	3.08	6.81	1.37	0.97	1.77	3.20	1.02
Salinas	OA	*mean*	90.49	79.62	84.37	89.46	88.61	80.15	88.88	86.26	85.17	88.80	87.91	87.62	85.92	**89.74**
		std	0.00	10.85	2.37	1.55	0.85	10.67	1.78	6.18	6.92	0.99	2.48	3.79	2.38	1.35
	AA	*mean*	94.72	81.24	89.78	93.79	93.47	82.64	93.52	90.39	88.39	93.23	92.86	92.07	90.89	**94.08**
		std	0.00	15.00	3.34	1.63	0.77	16.27	1.62	7.69	8.91	1.20	2.15	4.22	2.85	1.37
	Kappa	*mean*	89.67	78.20	83.17	88.57	87.66	78.78	87.95	85.20	84.02	87.87	86.93	86.62	84.81	**88.87**
		std	0.00	11.35	2.51	1.65	0.90	11.10	1.89	6.50	7.29	1.06	2.62	4.00	2.52	1.44

spectral graphs into SR, which clearly improves the classification performance when compared to GCSR [40] using only spectral graphs. However, its performance is still limited, which may be also attributed to the ranking-based selection strategy. It is believed that its performance can be further enhanced, if it employs spectral clustering to process SR coefficients for BS. GRSC [45] achieves relatively high performance, since it builds SR subspace clustering with graph learning between latent region features to preserve spatial structures, and conducts spectral clustering to select the final bands with the maximal IE in each cluster. In addition, S4P [46] also performs well, as it can learn the spatial relationship through predefined spatial graphs, and preserve the relationship consistency between the original space and the reconstructed one. However, the spectral relationship is only captured with SR between spectral bands in GRSC and S4P. Our method simultaneously models both spatial and spectral relationships and integrates them into an SR framework with consistency selection constraints. Thus, this approach allows to capture of complex patterns and relationships within the HSI, resulting in more effective BS and classification.

2) Average Performance of Different Number of Bands: Tables III and IV display the average OA, AA, and Kappa values across different numbers of selected bands for various BS methods on three datasets with KNN and SVM, respectively. The top-performing and second-best results are marked in bold and underlined, respectively.

The same conclusion can be drawn from Tables III and IV as Figs. 4 and 5 indicate, that the proposed method consistently outperforms the compared methods in terms of OA, AA, and Kappa values. The specific analysis is as follows.

1) The proposed method achieves the highest OA and Kappa values for both KNN and SVM classifiers across all three datasets. Specifically, the proposed method achieves OA improvements of at least 1.48%, 0.37%, and 0.28% for KNN on the three datasets, respectively. Similarly, the proposed method enhances OA for SVM by at least 1.64%, 0.08%, and 0.24% on the respective datasets. This indicates that the proposed method can effectively capture the main information and patterns of the dataset, achieving a high overall classification accuracy.
2) In terms of AA, the proposed method experiences a slight decrease only compared to GCSR-SC on Indian Pines. Specifically, the AA of the proposed method is 0.35% lower than GCSR-SC for KNN and is 0.18% lower for SVM. This may be due to specific challenges associated with Indian Pines, such as the presence of hard-to-classify categories or imbalanced distribution of samples across categories. Furthermore, the spectral clustering postprocessing employed by GCSR-SC may contribute to enhancing the discrimination power of the bands selected for classification. It is worth noting that, the proposed method demonstrates clear improvements in AA on the other two datasets. At Pavia University, it boosts AA by 0.36% and 0.13% for KNN and SVM, respectively. Similarly, on Salinas, it achieves AA improvements of 0.29% and 0.11% for KNN and SVM, respectively.
3) It can be observed that the standard deviation of OA, AA, and Kappa achieved by the proposed method across different numbers of selected bands is generally at a low level. The low standard deviation combined with the high average suggests that the method is robust to variations in the number of bands selected, as it

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TABLE IV

AVERAGE AND STANDARD DEVIATION OF EVALUATION METRICS OF SVM OVER DIFFERENT NUMBER (5:5:50) OF SELECTED BANDS

Dataset	Metrics (%)		All Bands	MVPCA	SMRS	E-FDPC	MDSR	GCSR	GCSR-SC	CDSP-MinV	RDGSR	GRSC	S4P	MCBS	SR-SSIM	Ours
Indian Pines	OA	*mean*	80.85	67.23	68.48	67.37	70.19	64.54	76.03	75.21	65.78	71.84	72.54	74.42	71.98	**77.67**
		std	0.00	7.53	5.71	4.52	4.03	7.52	4.87	8.98	10.11	6.02	5.75	10.01	7.35	4.08
	AA	*mean*	75.28	58.72	58.76	53.57	64.41	55.73	**69.32**	68.96	50.72	68.72	63.67	67.00	64.52	69.14
		std	0.00	11.72	7.57	5.78	5.47	11.53	7.52	12.78	15.00	8.58	8.35	15.45	9.86	6.20
	Kappa	*mean*	79.09	64.73	65.94	64.65	67.90	61.94	73.95	73.05	62.87	69.56	70.26	72.22	69.70	**75.69**
		std	0.00	8.09	6.12	4.86	4.32	8.00	5.18	9.62	10.93	6.48	6.17	10.74	7.82	4.32
Pavia University	OA	*mean*	94.40	85.34	90.56	91.34	90.56	83.50	91.42	90.64	87.34	92.12	92.49	91.03	89.29	**92.57**
		std	0.00	9.98	3.37	2.97	5.29	5.49	3.44	4.36	6.53	2.52	1.94	3.94	4.63	2.90
	AA	*mean*	92.70	76.92	87.70	88.66	87.14	78.47	88.66	87.10	82.01	89.50	89.89	87.59	85.54	**90.02**
		std	0.00	16.40	4.82	4.05	8.63	8.64	4.85	7.81	10.86	3.51	2.47	7.34	7.60	3.94
	Kappa	*mean*	92.73	81.50	87.89	88.85	87.89	79.07	88.96	87.97	83.87	89.85	90.31	88.45	86.29	**90.43**
		std	0.00	12.12	4.23	3.76	6.66	6.75	4.36	5.52	8.18	3.17	2.43	5.02	5.86	3.65
Salinas	OA	*mean*	93.12	87.70	90.31	92.96	92.66	88.26	92.83	91.54	90.69	92.88	92.71	92.06	92.03	**93.20**
		std	0.00	4.61	1.06	1.03	1.08	4.41	1.26	3.57	3.00	0.80	1.73	2.67	2.02	0.88
	AA	*mean*	96.62	91.42	94.67	96.39	96.21	91.32	96.30	94.86	94.45	96.19	96.23	95.70	95.58	**96.50**
		std	0.00	5.34	0.98	0.74	0.90	6.20	0.90	3.75	2.70	0.71	1.08	2.15	1.76	0.64
	Kappa	*mean*	92.50	86.69	89.47	92.31	91.99	87.29	92.17	90.79	89.86	92.23	92.04	91.35	91.31	**92.57**
		std	0.00	4.91	1.14	1.10	1.16	4.69	1.36	3.82	3.24	0.86	1.85	2.88	2.17	0.95

TABLE V

RUNNING TIME (IN SECONDS) OF DIFFERENT METHODS WHEN 30 BANDS ARE SELECTED

Dataset	MVPCA	SMRS	E-FDPC	MDSR	GCSR	GCSR-SC	CDSP-MinV	RDGSR	GRSC	S4P	MCBS	SR-SSIM	Ours
Indian Pines	0.038	1.861	0.038	0.047	0.069	0.089	0.033	3.269	3.637	0.483	0.193	3.359	0.245
Pavia University	0.199	6.706	0.103	0.068	0.188	0.228	0.070	15.675	3.468	1.445	0.914	14.916	1.300
Salinas	0.154	9.198	0.123	0.089	0.261	0.290	0.094	27.222[a]	5.396	1.391	0.646	8.931	1.351

[a] Due to encountering an out of memory issue, this time is test on another platform with 40GB memory and Intel i7-12700H.

maintains a high level of classification accuracy even when the number of bands changes. This suggests that the method is well-suited for HSI BS, where selecting an optimal number of bands can be a challenging task.

Overall, these results demonstrate the effectiveness of the proposed method in BS for HSI classification.

D. Computational Efficiency

To demonstrate the efficiency of different methods, a running time comparison among various methods by selecting 30 bands is conducted. It's worth noting that all algorithms are implemented on the same computing environment (MATLAB R2023a, an Intel i5-13400F CPU, and 32 GB of memory), ensuring the fairness of the test.

Table V presents the running time of different BS methods when selecting 30 bands. It can be observed that MVPCA, E-FDPC, GCSR, GCSR-SC, CDSP-MinV, and MCBS exhibit remarkable efficiency, completing the processing of three datasets in less than one second. On the other hand, SR-SSIM requires a longer runtime due to its need to compute pairwise SSIM for band ranking. Among the SR-based methods, MDSR stands out as the fastest, thanks to its strategy of sampling just dozens of pixels to represent each band, which significantly reduces the amount of data processed. GCSR and GCSR-SC only leverage spectral graphs with low computational complexity and have analytical solutions without requiring multiple iterations, which leads to short runtime. S4P also performs relatively quickly, as its modeling is constructed in the space of hundreds of superpixels, leading to efficient representation. GRSC is also modeled based on superpixels, but it requires computing Laplacian Eigenmaps for each superpixel, which results in slightly increased runtime. SMRS occupies more time due to the determination of the penalty parameter for the augmented Lagrange multiplier as described in [57]. RDGSR necessitates the construction of pixel-level graphs within each superpixel, whose computation complexity is about $\mathcal{O}(\sum_{i=1}^{N} M_i^2 B)$, resulting in increased runtime, particularly when dealing with larger superpixels.

Our method costs less runtime than SMRS, RDGSR, GRSC, S4P, and SR-SSIM. Specifically, our method has added the utilization of spatial relationships when compared to GCSR and GCSR-SC, while the increased runtime is acceptable. Unlike RDGSR, our method constructs a superpixel-level spatial graph with a computational complexity of $\mathcal{O}(N^2 B)$. Given that $N^2 \ll \sum_{i=1}^{N} M_i^2$, the runtime of our method is significantly reduced. In addition, since Pavia University and Salinas have significantly more pixels, the total runtime on them is also increased compared to that on Indian Pines.

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TABLE VI
COMPARISON OF AVERAGE OA BY SVM OF DIFFERENT METHODS WITH DIFFERENT BS STRATEGIES

Dataset	Selection Strategy	Spe	Spa	Direct SS	Joint SS	JSSGCSR	GCSR [40]	GRSC [45]
Indian Pines	Ranking	72.89	68.32	73.15	73.33	74.35	64.54	56.36
	Uniform Ranking	75.59	74.72	76.01	77.40	77.67	75.89	75.78
	Spectral Clustering	76.31	76.02	–	–	–	76.03	71.84
Pavia University	Ranking	89.06	90.95	89.81	89.06	91.58	83.50	81.18
	Uniform Ranking	92.25	91.55	92.33	92.28	92.57	91.82	91.76
	Spectral Clustering	91.96	91.75	–	–	–	91.42	92.12
Salinas	Ranking	92.90	92.22	92.93	92.89	92.86	88.26	88.66
	Uniform Ranking	92.96	92.96	93.10	93.11	93.20	92.85	92.81
	Spectral Clustering	93.12	92.97	–	–	–	92.83	92.88

Overall, based on Tables III–V, the proposed method demonstrates superior performance with high computational efficiency.

E. Ablation Experiment

In this section, the effectiveness of each module in the proposed method is investigated, including spatial relationships, spectral relationships, and consistent sparse SR. In addition, the effectiveness of the proposed uniform ranking-based BS strategy is explored.

Specifically, five different implementations based on SR are compared as follows.

1) *Spe:* Formulated as (8), which only integrates the spectral graph into the SR.
2) *Spa:* Formulated as (9), which only integrates the spatial graph into the SR.
3) *Direct SS:* Directly combines the results obtained from (8) and (9).
4) *Joint SS:* Formulated as (10), which jointly integrates both spectral and spatial graphs, while does not include a consistency constraint.
5) *JSSGCSR:* Formulated as (11), which jointly integrates both spectral and spatial graphs while also introducing a consistency constraint.

In terms of the BS strategy, the proposed uniform ranking is compared with traditional ℓ_2 norm-based ranking and spectral clustering. In order to comprehensively compare the three strategies, apart from the aforementioned five implementations, two additional methods, GCSR [40] and GRSC [45], are also employed to realize these three strategies.

The average OA of 5:5:50 selected bands achieved by the seven methods using the three strategies, evaluated by SVM is presented in Table VI. It is worth noting that spectral clustering is not applicable to three implementations including Direct SS, Joint SS, and JSSGCSR, because their coefficient matrices do not meet the requirement of being square.

1) Analysis of Five Implementations: By comparing the five implementations, we aim to assess the effectiveness of each module and the overall contribution of the proposed JSSGCSR.

1) The three methods that jointly integrate spatial and spectral graphs almost universally outperform those that only use either, regardless of the BS strategy employed. This underscores the importance of considering both spatial and spectral relationships in BS, especially when dealing with datasets with significant challenges.
2) The approach that solely relies on spatial graphs typically performs inferiorly compared to the method that solely utilizes spectral graphs, which suggests that spectral relationship among different bands plays a more crucial role in identifying key bands, while spatial relationships can complement the spectra' SR.
3) Among the three methods that jointly utilize spatial and spectral relationships, our proposed JSSGCSR achieves the best performance, indicating the importance and effectiveness of the joint consistency constraints. This demonstrates that by enforcing consistency between spatial and spectral information, our method is able to capture more representative bands, leading to improved classification accuracy.

2) Analysis of Three BS Strategies: By comparing the uniform ranking with traditional ranking and spectral clustering, we aim to assess its effectiveness in selecting bands for the SR-based methods.

1) Different BS strategies can significantly impact classification accuracy. The results presented in Table VI clearly demonstrate that spectral clustering can achieve the best performance, while the direct ranking exhibits the poorest performance. This may be due to that spectral clustering can capture complex relationships and patterns among bands, which are overlooked by direct ranking.
2) The proposed uniform ranking demonstrates a significant improvement over direct ranking, while its performance is slightly inferior to or comparable with spectral clustering. This is due to that uniform ranking can be viewed as a type of clustering, but it is an ordered and uniform clustering approach. Although it can reduce local correlations among bands, it still has limitations in terms of global decorrelation.
3) Although the proposed uniform ranking performs slightly worse than spectral clustering in terms of performance, it offers several advantages: a) lower computational complexity. Spectral clustering involves eigenvalue decomposition with a time complexity of $\mathcal{O}(B^3)$, whereas the computational complexity of uniform ranking is $\mathcal{O}(B^2)$. b) Wider applicability. Unlike spectral clustering, uniform ranking does not require the

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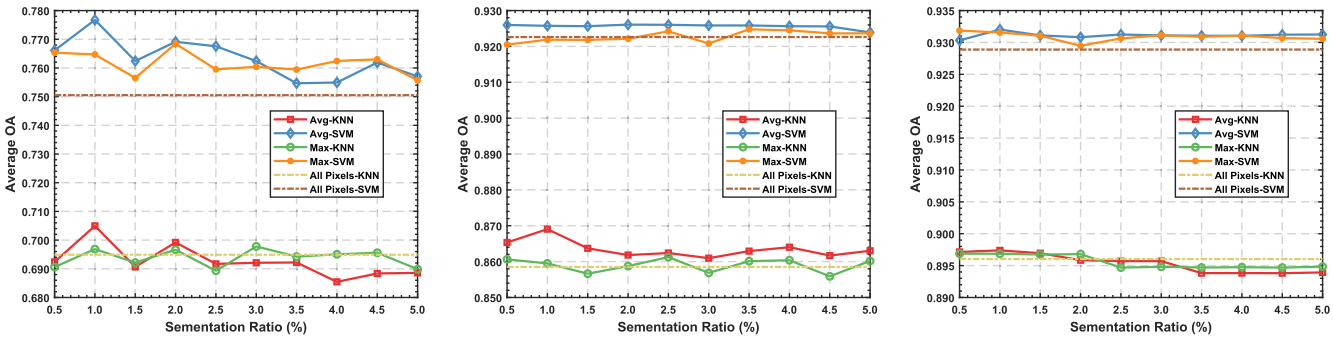


Fig. 6. Average OA under different spatial processing on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

input to be a square matrix. c) Simpler implementation with no parameters to tune.

F. Parameter Exploration

In this section, the impact of various parameters involved in the proposed method is explored. These parameters can be broadly categorized into two groups: the first group concerns the number of superpixels, as well as the pooling strategy in spatial segmentation and pooling; the second group involves the balancing parameters among different terms in the objective function.

1) Spatial Processing: The number of superpixels affects the granularity of spatial segmentation, and the pooling strategy determines how the information within each superpixel is aggregated. In this section, the proportion of the number of superpixels to that of original pixels is set within a range of {0.5%:0.5%:5%}. For spatial pooling, average pooling and max pooling are used to generate the superpixel representation. In addition, the processing without spatial segmentation and pooling, meaning building the spatial relationship graph between all original pixels, is also included in this comparison. The average OA of the proposed method under different numbers of superpixels and pooling strategies, as well as all pixels, is illustrated in Fig. 6.

From the figure, the utilization of superpixels and all original pixels are compared and analyzed as follows.

1) The utilization of all pixels and superpixels to construct spatial relationships and perform BS results in different classification performances, which vary across different datasets and classifiers. Overall, under the SVM classifier, the classification performance achieved through varying numbers of superpixels is generally superior to that obtained through all pixels, especially on the Indian Pines dataset. Under the KNN classifier, the difference in classification performance between using superpixels and processing all original pixels is less regular compared to that under the SVM classifier. Specifically, the classification performance achieved using all pixels may either exceed or be lower than that obtained using superpixels. However, when the number of superpixels is appropriately selected, using superpixels can achieve better classification performance.
2) In terms of computation, the time complexity of building spatial relationships using all pixels is about $\mathcal{O}(W^2H^2B + W^2H^2 \log WH)$. However, that of using N superpixels in our method, including PCA, superpixel segmentation, and spatial relationship construction, is about $\mathcal{O}(B^2WH + WH \log WH + N^2B + N^2 \log N)$. Since $WH \gg B$ and $WH \gg N$, adopting superpixels to build a spatial relationship is more computationally efficient. Furthermore, the space complexity of storing the distances between all pixels is $\mathcal{O}(W^2H^2)$, which can lead to high memory consumption issues for large datasets. For instance, Pavia University would require 320.5 GB memory for the $207\,400 \times 207\,400$ array in MATLAB. Although methods such as sparse matrix representation can reduce the demand for storage space, memory limitations still need to be taken into account. However, by setting the number of superpixels to 1% of the original pixels, this memory requirement can be significantly reduced to 0.032 GB.

Therefore, the superpixel-level spatial relationship can serve as an efficient alternative to pixel-level spatial relationships, making the utilization of spatial relationships more practical and feasible.

Furthermore, the comparative analysis of different pixel numbers and spatial pooling strategies is as follows.

1) The number of superpixels has varying impacts on the classification accuracy of selected bands across different datasets. Specifically, the Indian Pines exhibits a higher sensitivity to changes in superpixel numbers compared to the other two datasets, which could be attributed to the unique characteristics of Indian Pines, such as its lower spatial resolution, and more complex land cover patterns.
2) The number of superpixels has varying impacts on the classification performance of different classifiers. Overall, SVM appears to be less affected by changes in superpixel counts, while KNN tends to be slightly more sensitive to these changes. This observation can be attributed to the inherent characteristics of the two classifiers.
3) Theoretically, excessive superpixels lead to overly fragmented spatial relationships, whereas insufficient superpixels lead to overly coarse spatial relationships lacking details. Overall, in the Indian Pines, which is more sensitive to changes in superpixel counts, a segmentation ratio of 1.0% yields relatively superior results. This proportion holds true for the other two datasets as well.

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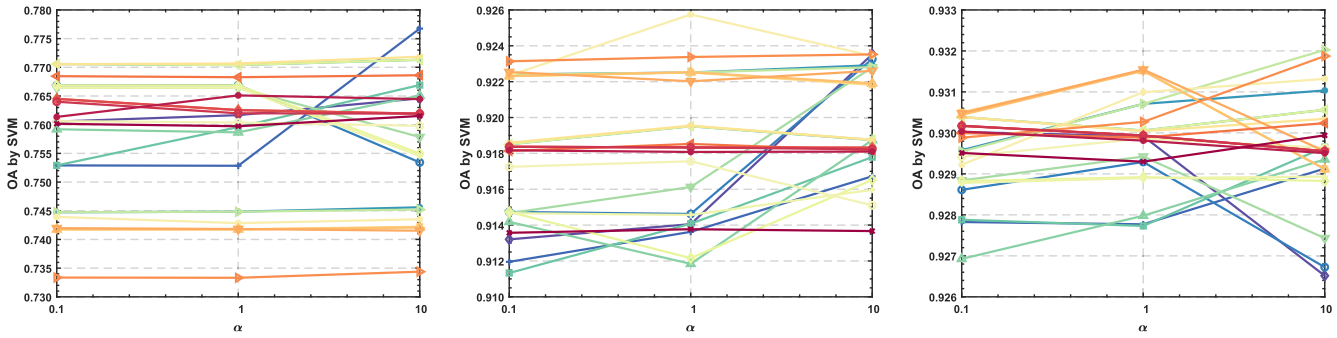


Fig. 7. Average OA by SVM under different α values and different β and γ combinations on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

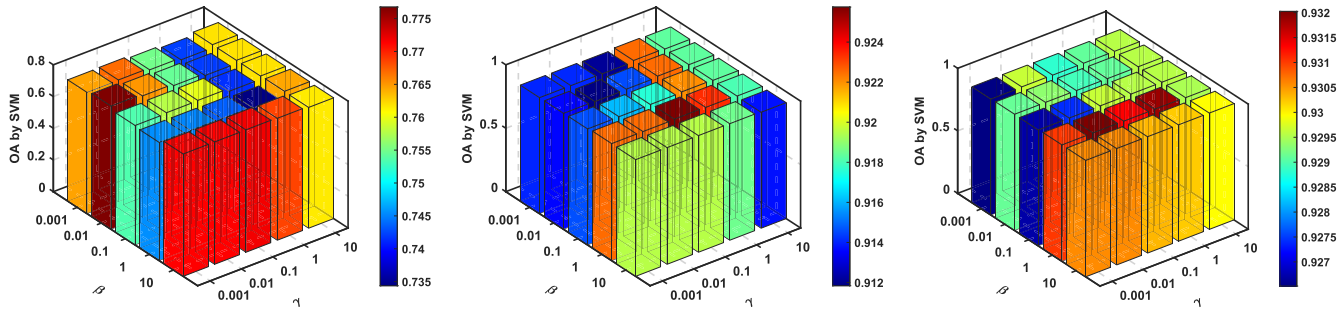


Fig. 8. Average OA by SVM under different β and γ values on (a) Indian Pines, (b) Pavia University, and (c) Salinas, respectively.

4) Overall, performing average pooling on superpixels tends to achieve better performance than max pooling. Max pooling mainly focuses on the maximum value within a region, which emphasizes prominent features but potentially leads to the loss of some spatial information. In contrast, average pooling as shown in Fig. 2 considers all pixels within a superpixel, which can smooth out the representation and better preserve spatial information. In addition, average pooling helps reduce noise, resulting in more robust superpixel features.

2) Model Hyperparameters: The objective function of JSSGCSR involves three hyperparameters, α, β, and γ. Among them, α is applied to the spatial graph SR error, while β and γ are imposed on different sparse regularization terms. In these experiments, the range of α is set to $\{0.1, 1, 10\}$, while the range β and γ is set to $\{0.001, 0.01, 0.1, 1, 10\}$.

In the first set of experiments, the impact of α is investigated. Fig. 7 illustrates the OA of SVM for different α values across 25 combinations of β and γ. It can be observed that the maximum difference in average OA for varying values of α is approximately 2% for Indian Pines, 1% for Pavia University, and 0.6% for Salinas. However, it must be noted that such differences only emerge for changes in α under a minority of combinations of β and γ. Overall, the impact of varying α on classification performance is relatively minor across most combinations of β and γ, which suggests that the proposed model is relatively robust to changes in α.

In the second set of experiments, the impact of β and γ is examined with α fixed. Fig. 8 illustrates the OA of SVM for different β and γ values. It can be observed from Fig. 8 that the impact of varying β and γ on the classification performance for the Indian Pines is slightly greater compared to their influence on the other two datasets. This indicates that the Indian Pines might be more responsive to alterations in the sparse regularization terms controlled by β and γ. Overall, a larger β is more suitable for all three datasets, since it can lead to more sparse discriminative power in both spatial and spectral SRs. The optimal value of γ may vary depending on the specific dataset. A larger γ leads to stronger consistency between the joint of spatial and spectral representations but may potentially limit the individual representational capabilities. Conversely, a smaller γ allows for more flexibility but may result in weaker consistency. Therefore, the optimal γ may be influenced by the spatial and spectral characteristics within the specific dataset.

V. Conclusion

This article proposes JSSGCSR to more effectively exploit spatial and spectral relationships for BS. JSSGCSR separately conducts SR to handle each view of the spatial and spectral graph and then enforces consistent sparse patterns on the joint spatial–spectral SR matrix to achieve a unified and robust representative band set. In addition, to reduce the heavy burden of building pixel-level spatial relationships, this article constructs spatial relationships at the superpixel level after applying spatial segmentation and pooling, aiming to reduce the burden of building spatial relationships. To optimize the proposed JSSGCSR model, the objective function is decomposed into two subproblems, and an efficient iterative algorithm with convergence is designed to obtain jointly consistent SR coefficients used for BS. In addition, the uniform ranking is designed to process the coefficients to achieve final BS results. Extensive experimental results on three benchmark datasets have clearly demonstrated the effectiveness of the proposed method.

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Despite the promising results, the proposed method has some limitations that need to be addressed in future work. First, the spatial and spectral relationship graphs in this article are both constructed based on the inherent characteristics of the HSI data and predefined models such as KNNs. Future work can integrate the utilization of prior information with graph learning algorithms to automatically explore the complex relationships in both spatial and spectral dimensions, thereby enhancing the performance of BS. Second, the number of superpixels for HSI segmentation is manually set, which may lead to inflexibility and suboptimal performance across different datasets. In the future, exploring adaptive methods for adjusting the number of superpixels could enhance the model's adaptability and performance in various scenarios.

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